Windows Server 2019

Domain Controller with RBAC

2000 Users — 5 Labs × 24 PCs — Lab Management API

Domain:	`emis.local`
Environment:	VMware ESXi 8.0
Prepared by:	EMIS IT Department
Date:	June 2026
Scripts:	4 (AD DS, RBAC, Users, Lab API)
Version:	2.0

Contents

Part 1: VM Sizing for Domain Controller

ESXi is assumed to be already installed and accessible via the web UI.

Resource	Value
vCPUs	2
RAM	4 GB (8 GB recommended)
Disk	60 GB thin provisioned
Network	1 vNIC (VM Network)
Guest OS	Windows Server 2019 Standard

Table 1: Domain Controller VM Specifications

Part 2: Download Windows Server 2019 ISO

- **Evaluation (180-day free trial):**
- `https://www.microsoft.com/en-us/evalcenter/evaluate-windows-server-2019`
- Choose **ISO** download, 64-bit edition (∼5.5 GB)

Part 3: Upload ISO and Create VM

3.1 Upload Windows Server ISO

1. In ESXi web UI → **Storage** → **Datastore browser**
2. Click **Create directory** → name it `ISOs`
3. Click **Upload** → select your Windows Server 2019 ISO
4. Wait for upload to complete (∼5.5 GB)

3.2 Create the Virtual Machine

1. **Virtual Machines** → **Create / Register VM**
2. Select **Create a new virtual machine** → Next
3. Configure:

Setting	Value
Name	`WinServer-DC`
Guest OS family	Windows
Guest OS version	Microsoft Windows Server 2019 (64-bit)
CPU	2
Memory	4096 MB (4 GB)
Hard disk 1	60 GB (Thin Provisioned)
Network Adapter	VM Network, VMXNET3
CD/DVD Drive	Datastore ISO file → Windows Server ISO

Table 2: VM Configuration

> ⚠ Important
>
> Check **"Connect at power on"** for the CD/DVD drive, otherwise the VM will not boot from the ISO.

Part 4: Install Windows Server 2019

4.1 Boot and Install

1. Power on the VM and open the console
2. Press any key when prompted to boot from CD/DVD
3. Set language, time, and keyboard → Click **Next** → **Install now**
4. Select: **Windows Server 2019 Standard (Desktop Experience)**

> ⚠ Important
>
> Always choose **Desktop Experience**. "Server Core" has no GUI and is command-line only.

5. Accept license terms → Next
6. Select **Custom: Install Windows only (advanced)**
7. Select **Drive 0 Unallocated Space** (60 GB) → Next
8. Wait 10–20 minutes for installation. VM reboots automatically.
9. Set **Administrator password** (e.g., `P@ssw0rd2024!`)

4.2 Configure Networking

1. Open **Network and Sharing Center** → **Change adapter settings**
2. Right-click **Ethernet0** → **Properties**

3. Select **Internet Protocol Version 4 (TCP/IPv4)** → Properties:

Setting	Value
IP address	`192.168.1.10`
Subnet mask	`255.255.255.0`
Default gateway	`192.168.1.1`
Preferred DNS	`127.0.0.1`
Alternate DNS	`8.8.8.8`

Table 3: Domain Controller IP Configuration

The Preferred DNS is set to `127.0.0.1` because this server will become the DNS server after AD DS promotion.

4.3 Set Computer Name

Open **PowerShell as Administrator** and run:

```
Rename-Computer -NewName "DC01" -Restart
```

Server reboots. Login again after reboot.

Part 5: Install Active Directory Domain Services

5.1 Copy Scripts to the Server

Option A — USB/Shared folder:
Copy the `ActiveDirectory` folder to `C:\Scripts\` on the server.

Option B — Download from GitHub:

```
Set-ExecutionPolicy RemoteSigned -Force

New-Item -Path "C:\Scripts" -ItemType Directory -Force

Invoke-WebRequest -Uri "https://raw.githubusercontent.com/
   sumanmhz/sumanmhz/main/ActiveDirectory/01-Install-AD-DS.ps1"
   -OutFile "C:\Scripts\01-Install-AD-DS.ps1"

Invoke-WebRequest -Uri "https://raw.githubusercontent.com/
   sumanmhz/sumanmhz/main/ActiveDirectory/02-Create-RBAC-
   Structure.ps1" -OutFile "C:\Scripts\02-Create-RBAC-Structure.
   ps1"
```

```
Invoke-WebRequest -Uri "https://raw.githubusercontent.com/
   sumanmhz/sumanmhz/main/ActiveDirectory/03-Bulk-Create-Users.
   ps1" -OutFile "C:\Scripts\03-Bulk-Create-Users.ps1"

Invoke-WebRequest -Uri "https://raw.githubusercontent.com/
   sumanmhz/sumanmhz/main/ActiveDirectory/04-AD-User-API.ps1" -
   OutFile "C:\Scripts\04-AD-User-API.ps1"
```

5.2 Step 1: Install AD DS and Promote to DC

```
cd C:\Scripts
.\01-Install-AD-DS.ps1
```

This script:

- Installs AD-DS, DNS, and GPMC server roles
- Promotes the server to a Domain Controller
- Creates the forest: `emis.local` (NetBIOS: `EMIS`)
- **Server reboots automatically** — wait 2–3 minutes

> **⚠ Important**
>
> After reboot, the login screen will show `EMIS\Administrator`. Use your Administrator password to login.

5.3 Step 2: Create RBAC Structure

```
cd C:\Scripts
.\02-Create-RBAC-Structure.ps1
```

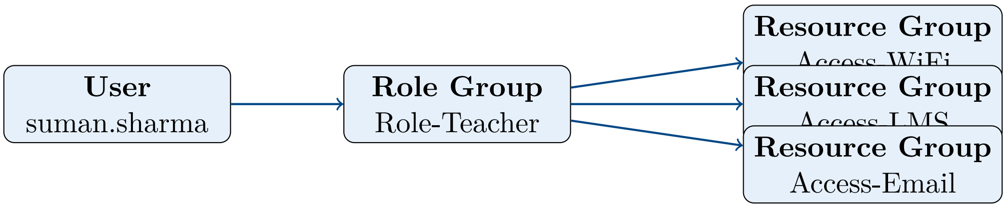
This script creates the complete RBAC framework:

Component	Count	Details
Top-level OUs	5	Users, Groups, Computers, Service Accounts, Disabled
Department OUs	10	IT, Admin, Faculty, Students, Finance, HR, Library, Research, Engineering, Management
Student sub-OUs	5	Batch-2080 through Batch-2084
Faculty sub-OUs	4	Computer, Electronics, Civil, Electrical Engineering
Role Groups	3	Global scope: Role-SuperAdmin, Role-Teacher, Role-Students
Resource Groups	11	DomainLocal scope (e.g., Access-WiFi, Access-LMS)
GPOs	4	Password policy, desktop config, lab computers, audit

Table 4: RBAC Structure Summary

5.3.1 RBAC Model

The access model follows industry best practice:



`User → Role Group → Resource Group → Actual Permission`

5.3.2 Role-to-Resource Mappings

All roles receive the same daily-use resources. Users experience the system normally — login with their AD username and password, access files, print, browse the internet, use email and LMS. No role feels "locked down." Only infrastructure access (server room, VPN, ERP) is limited to the superadmin role.

Role Group	Resource Access
Role-SuperAdmin	FileServer-RW, PrinterColor, PrinterBW, WiFi, VPN, ServerRoom, ERP, LMS, Email, InternetFull, LabComputers
Role-Teacher	FileServer-RW, PrinterColor, PrinterBW, WiFi, LMS, Email, InternetFull, LabComputers
Role-Students	FileServer-RW, PrinterColor, PrinterBW, WiFi, LMS, Email, InternetFull, LabComputers

Table 5: Role-Based Access Control Mappings

Teachers and students get the **exact same** resources. The only difference is what they can do through the **API** — teachers can monitor labs and reset passwords. On a PC, a student's experience is identical to a teacher's.

5.4 Step 3: Bulk Create 2000 Users

```
cd C:\Scripts
.\03-Bulk-Create-Users.ps1 -GenerateSample
```

- Enter a default password when prompted (e.g., `Welcome@123`)
- Generates 2000 users automatically:
  - 1400 students (distributed across batches)
  - 600 staff (across all departments)
- Each user is placed in the correct OU by department

- Each user is assigned their RBAC role group
- Password change is forced at first logon

To use your own user list instead of generated data, create a CSV file with columns: `FirstName, LastName, Username, Email, Department, Title, Role, Phone` and run:

```
.\03-Bulk-Create-Users.ps1 -CsvPath "C:\myusers.csv"
```

Part 6: Lab & User Management API (Step 4)

The API provides centralized management of AD users and lab PCs over REST. It uses **Pode** (pure PowerShell web framework) and **WinRM** (PowerShell Remoting) to control domain-joined PCs.

6.1 Three Roles

Role	Who	Permissions
`superadmin`	IT Admin	Full control: CRUD users, bulk create/delete by batch or faculty, shutdown/restart/install software on any PC in any lab
`teacher`	Teacher / TA	Monitor **any** lab (who's logged in, CPU/RAM, running apps), list students, reset student passwords. Teachers are not assigned to a fixed lab — they can walk into any lab and access its data.
`student`	Student	Change own password only

Table 6: API Role Definitions

6.2 Configure Labs

On first run, the script generates a template `labs.json`. Edit it to match your actual lab layout:

```
[
  {
    "Name": "Lab-1",
    "Location": "Block A, Room 101",
    "Switch": "10.10.1.1",
    "Subnet": "10.10.1.0/24",
    "Gateway": "10.10.1.1",
    "PCs": [
      { "Hostname": "LAB1-PC01", "IP": "10.10.1.11" },
```

```
        { "Hostname": "LAB1-PC02", "IP": "10.10.1.12" }
      ]
    }
]
```

Each lab has its own switch and subnet. PCs within a lab are interconnected through that switch.

6.3 Generate Service Keys

```
cd C:\Scripts

# Generate superadmin key
.\04-AD-User-API.ps1 -GenerateKey
# Select role: 1 (superadmin)
# Name: admin-portal

# Generate teacher key
.\04-AD-User-API.ps1 -GenerateKey
# Select role: 2 (teacher)
# Name: ram.sharma
# (no lab assignment - teachers can access any lab)

# Generate student key
.\04-AD-User-API.ps1 -GenerateKey
# Select role: 3 (student)
# Name: suman.adhikari
# AD username: suman.adhikari
```

△ Important

Copy each key immediately after generation. Keys are stored hashed and cannot be retrieved later. Keys are saved in `api-keys.json`.

6.4 Start the API Server

```
# HTTPS (production) - uses self-signed cert
.\04-AD-User-API.ps1

# HTTP mode (testing only)
.\04-AD-User-API.ps1 -HttpOnly -Port 8080

# Custom threads for high concurrency
.\04-AD-User-API.ps1 -Threads 32
```

The server runs on port `8443` (HTTPS) by default with 16 threads. This handles 120+ concurrent users easily.

6.5 API Endpoints

Method	Path	Role	Description
Users			
GET	`/api/v1/users`	super, teacher	List/search users
POST	`/api/v1/users`	super	Create user
POST	`/api/v1/users/bulk`	super	Bulk create (max 500)
DELETE	`/api/v1/users/{u}`	super	Disable user
DELETE	`/api/v1/users/batch/{b}`	super	Disable entire batch
DELETE	`/api/v1/users/faculty/{p}`	super	Disable entire program
Password			
PUT	`/api/v1/users/{u}/password`	all	Change password
POST	`/api/v1/users/{u}/reset`	super, teacher	Admin reset
Labs			
GET	`/api/v1/labs`	super, teacher	List labs
GET	`/api/v1/labs/{lab}`	super, teacher	Lab detail + PC status
GET	`/api/v1/labs/{lab}/monitor`	super, teacher	Live monitoring
PC Control			
POST	`/api/v1/pcs/{pc}/shutdown`	super	Shutdown PC
POST	`/api/v1/pcs/{pc}/restart`	super	Restart PC
POST	`/api/v1/pcs/{pc}/logoff`	super	Force logoff
POST	`/api/v1/pcs/{pc}/message`	super	Send popup message
POST	`/api/v1/labs/{lab}/shutdown-all`	super	Shutdown entire lab
POST	`/api/v1/labs/{lab}/restart-all`	super	Restart entire lab
POST	`/api/v1/labs/{lab}/message-all`	super	Broadcast message
Software			
GET	`/api/v1/pcs/{pc}/software`	super	List installed software
POST	`/api/v1/pcs/{pc}/install`	super	Install on one PC
POST	`/api/v1/labs/{lab}/install`	super	Install on entire lab

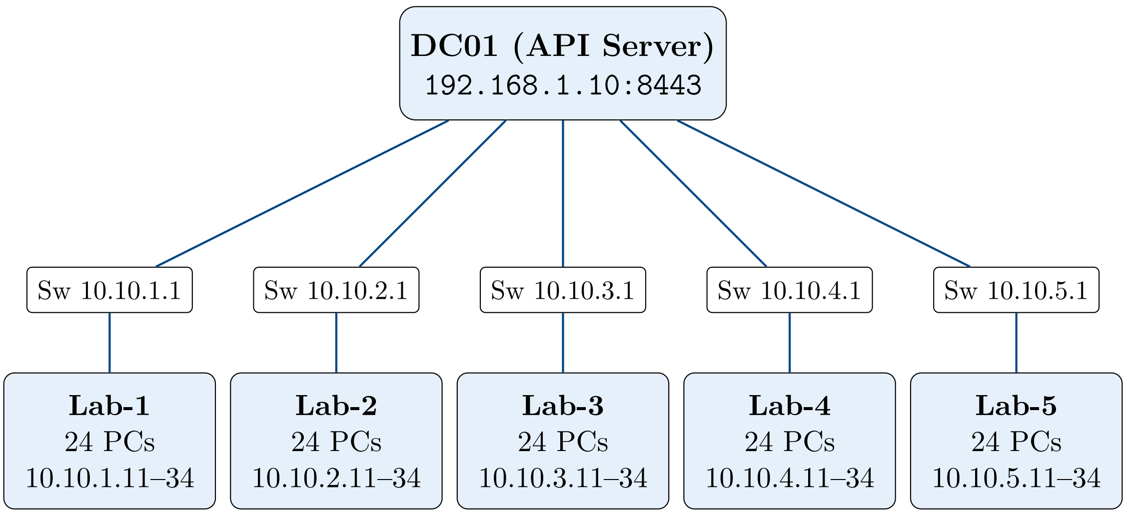
Table 7: API Endpoints Reference

6.6 Usage Examples

```
# Health check (no auth required)
curl https://dc:8443/api/v1/health

# List students in Batch-2082
curl -H "X-Api-Key: YOUR_KEY" \
  "https://dc:8443/api/v1/users?department=Students&batch=Batch
     -2082"

# Create a single user
```

```
curl -X POST -H "X-Api-Key: KEY" -H "Content-Type: application/
    json" \
  -d '{"firstName":"Ram","lastName":"Sharma","username":"ram.
     sharma","department":"Students","batch":"Batch-2083"}' \
  https://dc:8443/api/v1/users

# Monitor Lab-1 (who's logged in, CPU, RAM, processes)
curl -H "X-Api-Key: TEACHER_KEY" \
  https://dc:8443/api/v1/labs/Lab-1/monitor

# Install Firefox on all PCs in Lab-2
curl -X POST -H "X-Api-Key: ADMIN_KEY" \
  -H "Content-Type: application/json" \
  -d '{"wingetId":"Mozilla.Firefox"}' \
  https://dc:8443/api/v1/labs/Lab-2/install

# Shutdown all PCs in Lab-3
curl -X POST -H "X-Api-Key: ADMIN_KEY" \
  https://dc:8443/api/v1/labs/Lab-3/shutdown-all

# Student changes own password
curl -X PUT -H "X-Api-Key: STUDENT_KEY" \
  -H "Content-Type: application/json" \
  -d '{"currentPassword":"Welcome@123","newPassword":"MyN3w!
     Pass"}' \
  https://dc:8443/api/v1/users/suman.adhikari/password

# Disable entire graduated batch
curl -X DELETE -H "X-Api-Key: ADMIN_KEY" \
  https://dc:8443/api/v1/users/batch/Batch-2080
```

6.7 Lab Network Architecture



All lab PCs must be domain-joined and have **WinRM** enabled (enabled by default via Group Policy on domain-joined machines). The API uses PowerShell Remoting to execute commands on PCs remotely — no agent installation needed.

6.8 Enable WinRM on Lab PCs

WinRM is usually enabled automatically on domain-joined PCs. If not, run this on each PC (or push via GPO):

```
# On each lab PC (as admin)
Enable-PSRemoting -Force
Set-Item WSMan:\localhost\Client\TrustedHosts -Value "DC01" -
   Force
```

Or deploy via Group Policy:

1. Open **Group Policy Management** → create GPO under `EMIS Computers`
2. Navigate to: Computer Configuration → Policies → Administrative Templates → Windows Components → Windows Remote Management
3. Enable **Allow remote server management through WinRM**
4. Set IPv4 filter to `*`

Part 7: Lab PC Setup (120 Computers)

Every lab PC must be configured so that teachers and students can log in with their domain accounts, and the API server can manage them remotely.

△ Important

User data is isolated. Each user gets a separate profile folder on the PC (`C:\Users\username\`). Students cannot access each other's files. Only SuperAdmins (Domain Admins) can access all profiles. Files are stored locally on each PC's own disk — the server does **not** share RAM or storage with lab PCs.

7.1 What Each PC Needs

#	Requirement	Details
1	Windows 10/11 Pro or Education	Home edition cannot join a domain
2	Static IP address	Each PC gets a fixed IP from its lab subnet (e.g., `10.10.1.11`)
3	DNS pointing to DC	Primary DNS = `192.168.1.10` (the Domain Controller)
4	Domain-joined	Joined to `emis.local` domain
5	WinRM enabled	For remote management via the API
6	Firewall rules	Allow WinRM (TCP 5985/5986) and ICMP ping from DC

Table 8: Lab PC Requirements Checklist

⚠ Important

Windows Home edition cannot join a domain. All lab PCs must run Windows 10/11 **Pro**, **Enterprise**, or **Education**.

7.2 Step-by-Step PC Setup

Repeat for each of the 120 PCs (or use a Windows deployment tool like WDS/MDT for imaging):

7.2.1 1. Configure Static IP

Set the IP based on the lab subnet. Example for Lab-1, PC #5:

IP:	`10.10.1.15`
Subnet:	`255.255.255.0`
Gateway:	`10.10.1.1`
DNS 1:	`192.168.1.10` (DC01)
DNS 2:	`8.8.8.8` (fallback)

Or via PowerShell:

```
# Set static IP (adjust InterfaceIndex, IP, gateway per PC)
$nic = Get-NetAdapter | Where-Object Status -eq "Up"
New-NetIPAddress -InterfaceIndex $nic.ifIndex -IPAddress
   "10.10.1.15" `
    -PrefixLength 24 -DefaultGateway "10.10.1.1"
Set-DnsClientServerAddress -InterfaceIndex $nic.ifIndex `
    -ServerAddresses "192.168.1.10","8.8.8.8"
```

7.2.2 2. Join Domain

```
Add-Computer -DomainName "emis.local" -Restart
```

Enter `EMIS\Administrator` credentials when prompted. PC reboots and is now domain-joined.

7.2.3 3. Enable WinRM

WinRM is usually auto-enabled on domain-joined PCs via Group Policy. Verify:

```
# Check if WinRM is running
Get-Service WinRM

# If not enabled, run:
Enable-PSRemoting -Force
```

7.2.4 4. Set Computer Name

Use a consistent naming scheme: `LAB{lab#}-PC{number}`

```
# Example: Lab 2, PC 7
Rename-Computer -NewName "LAB2-PC07" -DomainCredential (Get-
    Credential) -Restart
```

7.2.5 5. Verify from DC

After setup, test from the Domain Controller:

```
# Ping
Test-Connection -ComputerName LAB1-PC01 -Count 2

# Test WinRM
Test-WSMan -ComputerName LAB1-PC01

# Run remote command
Invoke-Command -ComputerName LAB1-PC01 -ScriptBlock { hostname
    }
```

7.3 Bulk Setup with GPO

Instead of configuring 400 PCs manually, use Group Policy to push settings automatically:

GPO Setting	Path
Enable WinRM	Computer Config → Admin Templates → Windows Remote Management → Allow remote server management
Firewall: WinRM	Computer Config → Windows Settings → Security → Windows Firewall → Inbound Rules → Predefined: Windows Remote Management
Firewall: ICMP	Same path → Predefined: File and Printer Sharing (Echo Request)
Restrict local login	User Config → Admin Templates → Control Panel → limit Settings pages

Table 9: GPO Settings for Lab PCs

7.4 Windows Deployment (Optional)

For imaging 400 PCs efficiently:

1. Install **Windows Deployment Services (WDS)** on a server
2. Create a **reference image**: install Windows, join domain, enable WinRM, install standard software
3. Capture the image with **Microsoft Deployment Toolkit (MDT)**
4. PXE-boot each lab PC → select the image → auto-deploys in 15–20 minutes
5. Use an `unattend.xml` answer file to auto-set hostname, IP, and domain join

With MDT + WDS, you can deploy all 120 PCs in a single weekend. Each PC boots from the network, gets the image, joins the domain, and is ready for students and teachers.

7.5 What Teachers and Students Get

Once a PC is set up, users log in with their domain credentials:

Username: `EMIS\suman.adhikari`
Password: Their AD password (forced change on first login)

Feature	Teacher	Student
Login to any lab PC	✓	✓
Access shared drives (R/W)	✓	✓
Print (BW & Color)	✓	✓
Internet (full, unrestricted)	✓	✓
Email & LMS	✓	✓
Install software	— (IT installs via API)	—
API-only features (not visible on the PC):		
View lab PC status via API	✓(any lab)	—
Reset student passwords via API	✓	—
Change own password via API	✓	✓

Table 10: Teacher vs Student Capabilities on Lab PCs

On the PC itself, teachers and students have the **exact same experience**. They log in with their AD credentials and use the PC normally. The only difference is what they can do through the **API** — teachers can monitor labs and reset passwords, students can only change their own password.

Part 8: Verification

After running all three scripts, verify the deployment:

8.1 Check Users

```
# Total users
Get-ADUser -Filter * | Measure-Object

# Students
Get-ADUser -Filter * -SearchBase "OU=Students,OU=EMIS Users,DC=
   emis,DC=local" | Measure-Object

# List first 10
Get-ADUser -Filter * -SearchBase "OU=EMIS Users,DC=emis,DC=
   local" -Properties Department |
    Select-Object -First 10 Name, SamAccountName, Department |
    Format-Table -AutoSize
```

8.2 Check Groups

```
# List role groups
```

```
Get-ADGroup -Filter 'Name -like "Role-*"' | Select Name

# Count members
Get-ADGroupMember "Role-Students" | Measure-Object
Get-ADGroupMember "Role-Teacher" | Measure-Object
```

8.3 Check OUs

```
Get-ADOrganizationalUnit -Filter * |
    Select Name, DistinguishedName |
    Format-Table -AutoSize
```

8.4 GUI Tool

Open **Active Directory Users and Computers**:

```
dsa.msc
```

Part 9: Join Client PC to Domain (Optional)

To test with a Windows 10/11 client VM:

9.1 Create Client VM

Setting	Value
Name	Client-PC01
OS	Windows 10/11 64-bit
CPU	2
RAM	2048 MB
Disk	40 GB

9.2 Configure Client DNS

> △ Important
>
> The client's DNS **must** point to the Domain Controller's IP (`192.168.1.10`), not `8.8.8.8`. Without this, domain join will fail.

IP:	`192.168.1.20`
Subnet:	`255.255.255.0`
Gateway:	`192.168.1.1`
DNS:	`192.168.1.10`

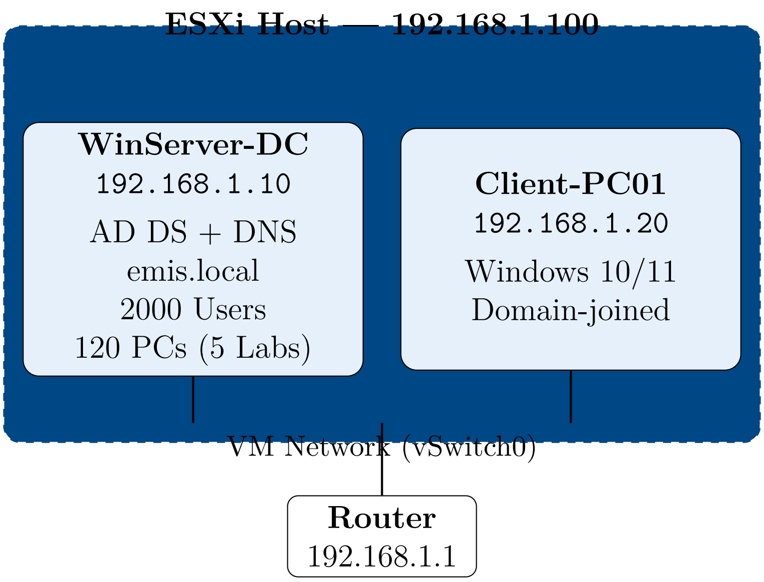
9.3 Join Domain

```
Add-Computer -DomainName "emis.local" -Restart
```

Enter domain admin credentials when prompted. After reboot, login as:

Username:	`EMIS\suman.sharma`
Password:	`Welcome@123` (default — will force change)

Part 10: Network Diagram



Part 11: Troubleshooting

Problem	Solution
VM can't boot from ISO	Check CD/DVD drive is connected and "Connect at power on" is checked
Script says "not recognized"	Run `Set-ExecutionPolicy RemoteSigned -Force` first

Problem	Solution
AD DS promotion fails	Ensure static IP and DNS (`127.0.0.1`) are set; server must ping the gateway
Client can't join domain	Client's DNS **must** point to DC's IP (`192.168.1.10`), not `8.8.8.8`
"Access Denied" running scripts	Right-click PowerShell → "Run as Administrator"
VM won't start (DVMT error)	Reduce video memory in VM settings to 8 MB
API won't start ("No API keys")	Run `.\04-AD-User-API.ps1 -GenerateKey` first to create at least one key
Pode not found	Run `Install-Module -Name Pode -Force -Scope CurrentUser`
PC shows offline in monitor	Check WinRM is enabled: `Test-WSMan -ComputerName LAB1-PC01`
Cannot remote to lab PC	Ensure PC is domain-joined and WinRM firewall rule is open (TCP 5985/5986)
API key rejected (403)	Verify the key in `api-keys.json` matches exactly; keys are case-sensitive

Table 11: Common Issues and Solutions

Part 12: Execution Summary

Order	Time	Action
1	10 min	Download Windows Server 2019 ISO, upload to ESXi datastore
2	5 min	Create VM (2 vCPU, 4 GB RAM, 60 GB disk) in ESXi
3	20 min	Install Windows Server 2019 Desktop Experience
4	5 min	Set static IP (192.168.1.10), DNS (127.0.0.1), hostname DC01
5	5 min	Run `01-Install-AD-DS.ps1` → server reboots
6	2 min	Run `02-Create-RBAC-Structure.ps1` → OUs, groups, GPOs
7	10 min	Run `03-Bulk-Create-Users.ps1 -GenerateSample` → 2000 users
8	5 min	Edit `labs.json` with actual lab IPs, switches, PC hostnames
9	2 min	Run `04-AD-User-API.ps1 -GenerateKey` → create service keys
10	1 min	Run `04-AD-User-API.ps1` → API starts on port 8443
11	5 min	Verify with `dsa.msc`, join client PCs, test API endpoints

Table 12: Step-by-Step Execution Order

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